the negative plates combine with sulphate molecules to form lead sulphate and hydrogen. The combination of lead peroxide and hydrogen at the positive electrode produces water and lead sulphate. The water dilutes the electrolyte, making it a weaker solution, drop in cell voltage, hence specific gravity of electrolyte is reduced.

During recharging, current is made to flow into the positive electrode of each cell. This current causes the lead sulphate at the negative electrode to recombine with hydrogen ions, thus re-forming sulphuric acid in the electrolyte and Spongy lead on the negative plates. Also, the lead sulphate on the positive electrodes recombines with water to regenerate lead peroxide on the positive plates and sulphuric acid in the electrolyte. Thus cell voltage is increased and the specific gravity of electrolyte is increased.

The two way of reversible chemical reaction (charged on the left and discharged on the right) is given as

Lead Peroxide + Sulphuric Acid + Spongy Lead ⇆ Lead Sulphate + Water + Electrolyte

$PbO_2 + 2H_2SO_4 + Pb \leftrightarrows PbSO_4 + 2H_2O + PbSO_4$

10.1.3 Battery Charging

The battery is charged by the vehicle generator during vehicle running. When the engine is in off condition, the electrical device will utilise the stored electrical energy from the battery. This will quickly discharge the battery and it requires external charging to charge the battery.

Method of Charging

1. Constant Voltage Method
2. Constant Current Method
3. Quick Charging Method

1. Constant voltage method

The charging voltage is kept constant throughout the charging process. The charging current is high in the begining when the battery is in the discharge condition. The current is gradually dropping off as the battery picks up charge

2. Constant current method

The charging current is kept constant throughout the charging period. The charging may be carried out in two steps. An initial charging of approximately higher volt and a finishing rate of low volt is used to avoid excessive gassing or overheating.

3. Quick charging method

The 80% of the total charging is done with high current and later the charging current is reduced. A battery in good condition alone charged in this method.

Specific gravity	Cell Voltage	Battery Voltage	% Charge
1.28	2.12 V	12.7 V	100
1.24	2.08 V	12.5 V	70
1.20	2.04 V	12.3 V	50
1.15	1.99 V	12.0 V	20
1.12	1.96 V	11.8 V	0

A battery is called as 'dead battery' when the battery is completely discharged condition.

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